

Type-B Self-Similarity of the Fake-Degree Frobenius Sum at a Root of Unity: The Even- e Case

Clio

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Abstract

Let $W_B = \mathbb{Z}/2 \wr S_n$ be the hyperoctahedral group with reflection representation $V \cong \mathbb{C}^n$ and degrees $\{2, 4, \dots, 2n\}$. For a signed cycle type (α, β) (with $|\alpha| + |\beta| = n$) put

$$\Psi_{(\alpha, \beta)}^B(t) := \frac{\prod_{i=1}^n (1 - t^{2i})}{\prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b)} \in \mathbb{Z}[t],$$

$z_{(\alpha, \beta)}^B := 2^{\ell(\alpha) + \ell(\beta)} z_\alpha z_\beta$, and, for ζ_e a primitive e -th root of unity,

$$q_e^{B, (n)} := \sum_{|\alpha| + |\beta| = n} \frac{\Psi_{(\alpha, \beta)}^B(\zeta_e)}{z_{(\alpha, \beta)}^B} p_\alpha(x) p_\beta(y) \in \Lambda(x) \otimes \Lambda(y) \otimes \mathbb{Z}[\zeta_e].$$

Theorem. *For every even $e \geq 2$ and every $n \geq 0$, with $k' := \lfloor n/e \rfloor$ and $r_0 := n \bmod e$,*

$$\boxed{q_e^{B, (n)} = p_{(e/2, e/2)}(y)^{k'} \cdot q_e^{B, (r_0)}.$$

The proof is a companion to the odd- e theorem [4]: the positive alphabet x is now inert, and the entire self-similarity factor lives on the negative alphabet y . Two orthogonal cancellations balance a twin double-zero at $\zeta_e^{e/2} = -1$: a numerator zero of order $2k' + \varepsilon$ from the “over-generation” factor $\prod_m (1 - t^{em})$ meets a denominator zero of the same order from $(1 + t^{e/2})^{2k' + \varepsilon}$, where $\varepsilon := \lfloor r_0 \geq e/2 \rfloor$. The L’Hôpital limit contributes $2^{2k' + \varepsilon} (2k' + \varepsilon)!$, and the “non-collapsed” numerator residues fold via the reindexing $\phi(s) = 2s \bmod e$ (now 2-to-1 instead of 1-to-1) into $(e/2)^{2k'}$ factors — each contributed by Lemma 2 applied to the primitive $(e/2)$ -th root of unity ζ_e^2 . Together with the centraliser factor $e^{2k' + \varepsilon} (2k' + \varepsilon)!$, everything cancels exactly and the coefficient identity $c_{\alpha, ((e/2)2k', \beta_{r_0})}^{B, (n)} = c_{\alpha, \beta_{r_0}}^{B, (r_0)}$ holds as literal equality in $\mathbb{Z}[\zeta_e]$.

This is the fifth theorem of the composite- d cycle. Combined with the odd- e result [4], it completes the type-B self-similarity story: for every $e \geq 2$, the “residue factor” p_e^B is a single power sum of a single alphabet — $p_{(e)}(x)$ for odd e , $p_{(e/2, e/2)}(y)$ for even e . The parity dichotomy is genuine and arithmetic (Section 8), and both cases are established without invocation of the Verschiebung machinery projected by the morning dream of 2026-08-06.

1 Setup

Fix even $e \geq 2$ and let $\zeta_e := e^{2\pi i/e}$. Throughout, $n \geq 0$, $k' := \lfloor n/e \rfloor$, and $r_0 := n \bmod e \in \{0, 1, \dots, e-1\}$. Set

$$\varepsilon := \lfloor r_0 \geq e/2 \rfloor \in \{0, 1\}, \quad M := 2k' + \varepsilon,$$

so that $M = \lfloor 2n/e \rfloor$; and for $s \in \{1, 2, \dots, e/2 - 1\}$ put

$$h_s(r_0) := [s \leq r_0] + [s + e/2 \leq r_0] \in \{0, 1, 2\}.$$

Bipartitions of n index both the conjugacy classes and the irreducible characters of W_B . For a class of signed cycle type (α, β) with $|\alpha| + |\beta| = n$, the characteristic polynomial of a representative $w \in W_B$ acting on V is $\prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b)$, and by Chevalley–Molien for the reflection group W_B ([8, §3.20]) the graded character of the W_B -coinvariant algebra at w is

$$\Psi_{(\alpha, \beta)}^B(t) = \frac{\prod_{i=1}^n (1 - t^{2i})}{\prod_{a \in \alpha} (1 - t^a) \prod_{b \in \beta} (1 + t^b)}, \quad (1)$$

a polynomial in $\mathbb{Z}[t]$. The centraliser $|Z_{W_B}(w)| = z_{(\alpha, \beta)}^B := 2^{\ell(\alpha) + \ell(\beta)} z_\alpha z_\beta$, where $z_\gamma := \prod_i i^{m_i(\gamma)} m_i(\gamma)!$. Writing $q_e^{B, (n)} = \sum_{(\alpha, \beta)} c_{\alpha, \beta}^{B, (n)} p_\alpha(x) p_\beta(y)$, the coefficients are

$$c_{\alpha, \beta}^{B, (n)} = \frac{\Psi_{(\alpha, \beta)}^B(\zeta_e)}{z_{(\alpha, \beta)}^B} \in \mathbb{Z}[\zeta_e]. \quad (2)$$

2 Cyclotomic and Arithmetic Lemmas

Lemma 1 (Classical cyclotomic identity, primitive e -th roots). $\prod_{r=1}^{e-1} (1 - \zeta_e^r) = e$.

Proof. Setting $x = 1$ in the factorisation $x^e - 1 = (x - 1) \prod_{r=1}^{e-1} (x - \zeta_e^r)$ and taking the limit gives $\prod_{r=1}^{e-1} (1 - \zeta_e^r) = \lim_{x \rightarrow 1} (x^e - 1)/(x - 1) = e$. $\square$

Lemma 2 (Secondary cyclotomic identity, primitive $(e/2)$ -th roots). $\prod_{s=1}^{e/2-1} (1 - \zeta_e^{2s}) = e/2$.

Proof. Since $\zeta_e^2 = e^{2\pi i \cdot 2/e} = e^{2\pi i/(e/2)}$ is a primitive $(e/2)$ -th root of unity, Lemma 1 applied at $e' = e/2$ (in the variable $\eta = \zeta_e^2$) gives $\prod_{s=1}^{e/2-1} (1 - \eta^s) = e/2$. $\square$

Lemma 3 (Twin double-zero L'Hôpital). *For every integer $M \geq 0$,*

$$\lim_{t \rightarrow \zeta_e} \frac{\prod_{m=1}^M (1 - t^{em})}{(1 + t^{e/2})^M} = 2^M M!.$$

Proof. Substitute $v = -t^{e/2}$; then $t \rightarrow \zeta_e$ corresponds to $v \rightarrow 1$ (since $\zeta_e^{e/2} = -1$). Because $2m$ is even, $t^{em} = (t^{e/2})^{2m} = (-v)^{2m} = v^{2m}$; and $1 + t^{e/2} = 1 - v$. Hence

$$\prod_{m=1}^M \frac{1 - t^{em}}{1 + t^{e/2}} = \prod_{m=1}^M \frac{1 - v^{2m}}{1 - v}.$$

Taking $v \rightarrow 1$ termwise, each factor tends to $2m$, so the product tends to $\prod_{m=1}^M 2m = 2^M M!$. $\square$

Lemma 4 (Non-vanishing of small tags). *For every integer j with $1 \leq j \leq e/2 - 1$, both $1 - \zeta_e^j$ and $1 + \zeta_e^j$ are nonzero elements of $\mathbb{Z}[\zeta_e]$.*

Proof. $1 - \zeta_e^j = 0 \iff e \mid j$; ruled out by $j < e/2 < e$. And $1 + \zeta_e^j = 0 \iff \zeta_e^j = -1 = \zeta_e^{e/2} \iff j \equiv e/2 \pmod{e}$; ruled out by $j < e/2$. $\square$

3 The Coefficient Formula

We record the target support shape upfront; it will be justified in Section 4.

Definition 5 (Even- e shape). A pair (α, β) of partitions with $|\alpha| + |\beta| = n$ has *even- e shape* if

- every part of α lies in $\{1, 2, \dots, e/2 - 1\}$;
- β has the form $\beta = ((e/2)^M, \beta')$, where β' is a partition all of whose parts lie in $\{1, 2, \dots, e/2 - 1\}$;
- $|\alpha| + |\beta'| = r_0 - (e/2)\varepsilon$.

Note in particular that $r_0 - (e/2)\varepsilon \in \{0, 1, \dots, e/2 - 1\}$ in both parity cases of ε , so shape pairs are consistent for every $n \geq 0$.

Proposition 6 (Closed-form coefficient). *For even $e \geq 2$ and every even- e shape pair (α, β) with $\beta = ((e/2)^M, \beta')$,*

$$c_{\alpha, \beta}^{B, (n)} = \frac{(2/e)^\varepsilon \prod_{s=1}^{e/2-1} (1 - \zeta_e^{2s})^{h_s(r_0)}}{z_{(\alpha, \beta')}^B \cdot \prod_{a \in \alpha} (1 - \zeta_e^a) \cdot \prod_{b \in \beta'} (1 + \zeta_e^b)} \in \mathbb{Z}[\zeta_e]. \quad (3)$$

The right-hand side is a nonzero element of $\mathbb{Z}[\zeta_e]$: the denominator is nonzero by Lemma 4 and $z^B > 0$; the numerator is a nonzero product of factors $1 - \zeta_e^{2s}$ (each nonzero by Lemma 4, applied to $j = 2s$ when $2s \neq e/2$, or via $1 - \zeta_e^{e/2} = 2 \neq 0$ when $s = e/4$).

Proof. Split the numerator of $\Psi_{(\alpha, \beta)}^B$ by whether $e/2 \mid i$:

$$\prod_{i=1}^n (1 - t^{2i}) = \underbrace{\prod_{m=1}^M (1 - t^{em})}_{=: A_1(t)} \cdot \underbrace{\prod_{\substack{i \in [1, n] \\ (e/2) \nmid i}} (1 - t^{2i})}_{=: A_2(t)}, \quad (4)$$

using that the set $\{i \in [1, n] : (e/2) \mid i\}$ is exactly $\{(e/2), (e/2) \cdot 2, \dots, (e/2) \cdot M\}$ (so $M = \lfloor 2n/e \rfloor$ as claimed), and that when $i = (e/2)m$, $t^{2i} = t^{em}$.

Split the denominator using $\beta = ((e/2)^M, \beta')$ and $\ell((e/2)^M) = M$:

$$\prod_{b \in \beta} (1 + t^b) = (1 + t^{e/2})^M \cdot \prod_{b \in \beta'} (1 + t^b); \quad (5)$$

by hypothesis α 's parts lie in $\{1, \dots, e/2 - 1\}$, so all factors $(1 - \zeta_e^a)$ (for $a \in \alpha$) and $(1 + \zeta_e^b)$ (for $b \in \beta'$) are nonzero at $t = \zeta_e$ by Lemma 4.

Assemble:

$$\Psi_{(\alpha, \beta)}^B(\zeta_e) = \left(\lim_{t \rightarrow \zeta_e} \frac{A_1(t)}{(1 + t^{e/2})^M} \right) \cdot \frac{A_2(\zeta_e)}{\prod_{a \in \alpha} (1 - \zeta_e^a) \prod_{b \in \beta'} (1 + \zeta_e^b)}. \quad (6)$$

The limit equals $2^M M!$ by Lemma 3.

Evaluate $A_2(\zeta_e)$. As i runs over $\{i \in [1, n] : (e/2) \nmid i\}$, its residue class $i \bmod e$ takes each value in $\{1, 2, \dots, e-1\} \setminus \{e/2\}$, appearing $k' + [i \bmod e \leq r_0]$ times. The reindexing $\phi(i) := 2i \bmod e$ maps this domain onto the nonzero even residues $\{2, 4, \dots, e-2\}$, and is 2-to-1: for each $r' = 2s$ with

$s \in \{1, \dots, e/2 - 1\}$, the preimages modulo e are $\{s, s + e/2\}$ (both lying in $\{1, \dots, e - 1\} \setminus \{e/2\}$). Grouping by $r' = 2s$:

$$\begin{aligned} A_2(\zeta_e) &= \prod_{s=1}^{e/2-1} (1 - \zeta_e^{2s})^{(k' + [s \leq r_0]) + (k' + [s + e/2 \leq r_0])} \\ &= \left(\prod_{s=1}^{e/2-1} (1 - \zeta_e^{2s}) \right)^{2k'} \cdot \prod_{s=1}^{e/2-1} (1 - \zeta_e^{2s})^{h_s(r_0)} \\ &= (e/2)^{2k'} \prod_{s=1}^{e/2-1} (1 - \zeta_e^{2s})^{h_s(r_0)}, \end{aligned}$$

where the last step is Lemma 2. Substitute into (6):

$$\Psi_{(\alpha, \beta)}^B(\zeta_e) = 2^M M! \cdot (e/2)^{2k'} \cdot \frac{\prod_{s=1}^{e/2-1} (1 - \zeta_e^{2s})^{h_s(r_0)}}{\prod_{a \in \alpha} (1 - \zeta_e^a) \prod_{b \in \beta'} (1 + \zeta_e^b)}. \quad (7)$$

Now compute $z_{(\alpha, \beta)}^B$. Since $\beta = ((e/2)^M, \beta')$ with β' having no parts equal to $e/2$, we have $z_\beta = (e/2)^M M! z_{\beta'}$ and $\ell(\beta) = M + \ell(\beta')$. Hence

$$\begin{aligned} z_{(\alpha, \beta)}^B &= 2^{\ell(\alpha) + M + \ell(\beta')} z_\alpha (e/2)^M M! z_{\beta'} \\ &= 2^M (e/2)^M M! \cdot 2^{\ell(\alpha) + \ell(\beta')} z_\alpha z_{\beta'} \\ &= e^M M! \cdot z_{(\alpha, \beta')}^B. \end{aligned}$$

Dividing (7) by this:

$$c_{\alpha, \beta}^{B, (n)} = \frac{2^M M! (e/2)^{2k'}}{e^M M!} \cdot \frac{\prod_s (1 - \zeta_e^{2s})^{h_s(r_0)}}{z_{(\alpha, \beta')}^B \prod_a (1 - \zeta_e^a) \prod_{b \in \beta'} (1 + \zeta_e^b)}.$$

The prefactor simplifies as

$$\frac{2^M (e/2)^{2k'}}{e^M} = \left(\frac{2}{e}\right)^M \left(\frac{e}{2}\right)^{2k'} = \left(\frac{2}{e}\right)^{2k' + \varepsilon} \left(\frac{e}{2}\right)^{2k'} = \left(\frac{2}{e}\right)^\varepsilon,$$

proving (3).

For membership in $\mathbb{Z}[\zeta_e]$: recall $c_{\alpha, \beta}^{B, (n)}$ is defined via evaluation of the *polynomial* $\Psi_{(\alpha, \beta)}^B(t) \in \mathbb{Z}[t]$ at $t = \zeta_e$, divided by the positive integer $z_{(\alpha, \beta)}^B$; and $\Psi^B(\zeta_e)/z^B$ is by construction a coefficient of $q_e^{B, (n)}$ in the $\{p_\alpha(x)p_\beta(y)\}$ -basis. That $q_e^{B, (n)}$ has coefficients in $\mathbb{Z}[\zeta_e]$ follows from (2); the fact that our formula (3) equals this coefficient in $\mathbb{C}$ is verified by the computation just given. $\square$

4 Support Theorem

Theorem 7 (Support of $q_e^{B, (n)}$, even e). *For even $e \geq 2$ and $n \geq 0$, $c_{\alpha, \beta}^{B, (n)} \neq 0$ if and only if (α, β) has even- e shape (Definition 5). In particular, $k_e(\alpha) := |\{a \in \alpha : e \mid a\}| = 0$ for every (α, β) in the support.*

Proof. Since $\Psi_{(\alpha,\beta)}^B(t) \in \mathbb{Z}[t]$, $\Psi_{(\alpha,\beta)}^B(\zeta_e) = 0$ iff the rational function on the right of (1) has a strictly higher-order zero in the numerator than in the denominator at $t = \zeta_e$. Compute the orders.

Numerator order. $1 - \zeta_e^{2i} = 0 \iff e \mid 2i \iff (e/2) \mid i$. Hence

$$\text{ord}_{\zeta_e} \prod_{i=1}^n (1 - t^{2i}) = |\{i \in [1, n] : (e/2) \mid i\}| = \lfloor 2n/e \rfloor = M.$$

Denominator (positive) order. $1 - \zeta_e^a = 0 \iff e \mid a$, so this order equals $k_e(\alpha)$.

Denominator (negative) order. $1 + \zeta_e^b = 0 \iff \zeta_e^b = -1 = \zeta_e^{e/2} \iff b \equiv e/2 \pmod{e}$. Write $k_{e/2}^{\text{odd}}(\beta) := |\{b \in \beta : b \equiv e/2 \pmod{e}\}|$; then the negative denominator order is $k_{e/2}^{\text{odd}}(\beta)$.

Hence $\Psi_{(\alpha,\beta)}^B(\zeta_e) \neq 0$ iff

$$k_e(\alpha) + k_{e/2}^{\text{odd}}(\beta) = M. \quad (8)$$

(“ $\geq$ ” would violate polynomialhood; “ $=$ ” is the only possibility because the total order of the polynomial ratio is ≥ 0 .)

Mass bound. Every part of α divisible by e has size $\geq e$; every part of β congruent to $e/2 \pmod{e}$ has size $\geq e/2$. So

$$e k_e(\alpha) + (e/2) k_{e/2}^{\text{odd}}(\beta) \leq |\alpha| + |\beta| = n. \quad (9)$$

Divide (9) by $e/2$: $2k_e(\alpha) + k_{e/2}^{\text{odd}}(\beta) \leq 2n/e$. Subtract (8):

$$k_e(\alpha) \leq \frac{2n}{e} - M = \frac{2n}{e} - \lfloor 2n/e \rfloor < 1.$$

Since $k_e(\alpha)$ is a nonnegative integer, $k_e(\alpha) = 0$. Then (8) forces $k_{e/2}^{\text{odd}}(\beta) = M = 2k' + \varepsilon$.

Forcing part sizes. With $k_e(\alpha) = 0$ and $k_{e/2}^{\text{odd}}(\beta) = 2k' + \varepsilon$, the mass bound (9) becomes $|\beta| \geq (e/2)(2k' + \varepsilon) = k'e + (e/2)\varepsilon$; combined with $|\alpha| + |\beta| = n = k'e + r_0$,

$$|\alpha| \leq r_0 - (e/2)\varepsilon < e/2. \quad (10)$$

Every part of α is $\leq |\alpha| < e/2$, giving the first bullet of Definition 5. Suppose any part of β congruent to $e/2 \pmod{e}$ has size $\geq 3e/2$: then $|\beta| \geq (e/2)(2k' + \varepsilon - 1) + 3e/2 = k'e + (e/2)\varepsilon + e$, so $|\alpha| \leq r_0 - (e/2)\varepsilon - e < 0$, contradiction. Hence all $2k' + \varepsilon$ of these parts equal $e/2$: β contains $((e/2)^{2k'+\varepsilon})$ as a sub-multiset. Similarly, any part of β with $b \not\equiv e/2 \pmod{e}$ and $b \geq e$ would force $|\alpha| < 0$, so the remaining parts of β all lie in $\{1, \dots, e-1\} \setminus \{e/2\}$. Since their total mass is at most $|\alpha| + |\beta'| = r_0 - (e/2)\varepsilon < e/2$, each such part has size $< e/2$, giving the second and third bullets. This proves the “only if” direction.

Conversely, any pair (α, β) of even- e shape has $k_e(\alpha) = 0$ (parts of α are $< e/2 < e$) and $k_{e/2}^{\text{odd}}(\beta) = 2k' + \varepsilon$ (parts of β congruent to $e/2 \pmod{e}$: the $2k' + \varepsilon$ copies of $e/2$, and no others since β' has parts $< e/2$). Then (8) holds with equality, so $\Psi_{(\alpha,\beta)}^B(\zeta_e) \neq 0$; equivalently, the closed form (3) exhibits $c_{\alpha,\beta}^{B,(n)}$ as a nonzero element of $\mathbb{Z}[\zeta_e]$. $\square$

5 Main Theorem

Theorem 8 (Type-B self-similarity, even e). *For every even $e \geq 2$ and every $n \geq 0$, with $k' = \lfloor n/e \rfloor$ and $r_0 = n \bmod e$,*

$$q_e^{B,(n)} = p_{(e/2, e/2)}(y)^{k'} \cdot q_e^{B,(r_0)}.$$

Equivalently, $p_e^B := p_{(e/2, e/2)}(y)$, and the “residue factor” $F_e^{B, (n)} := q_e^{B, (n)} / p_{(e/2, e/2)}(y)^{k'}$ satisfies $F_e^{B, (n)} = q_e^{B, (r_0)}$; in particular it depends on n only through r_0 .

Proof. The key observation is that the closed-form coefficient (3) depends on $(\alpha, \beta') = (\alpha, \beta \setminus ((e/2)^M))$ and on r_0 , but is *independent* of k' .

Concretely, apply Proposition 6 both at parameter n and at parameter r_0 ; note $\lfloor r_0/e \rfloor = 0$ (since $0 \leq r_0 < e$), so at parameter r_0 the exponent of $e/2$ in β is $M_{\text{base}} = 0 + \varepsilon = \varepsilon$, matching the “base” shape $\beta_{r_0} = ((e/2)^\varepsilon, \beta')$. The values ε and $h_s(r_0)$ depend only on r_0 , so (3) at parameter r_0 reads

$$c_{\alpha, \beta_{r_0}}^{B, (r_0)} = \frac{(2/e)^\varepsilon \prod_{s=1}^{e/2-1} (1 - \zeta_e^{2s})^{h_s(r_0)}}{z_{(\alpha, \beta')}^B \prod_{a \in \alpha} (1 - \zeta_e^a) \prod_{b \in \beta'} (1 + \zeta_e^b)}, \quad (11)$$

where $\beta_{r_0} = ((e/2)^\varepsilon, \beta')$. Comparing with (3):

$$c_{\alpha, ((e/2)^M, \beta')}^{B, (n)} = c_{\alpha, ((e/2)^\varepsilon, \beta')}^{B, (r_0)} \quad (\text{literal equality in } \mathbb{Z}[\zeta_e]). \quad (12)$$

Every element of the support of $q_e^{B, (n)}$ has the form $(\alpha, \beta) = (\alpha, ((e/2)^M, \beta')) = (\alpha, (e/2)^{2k'} \cup \beta_{r_0})$ where $\beta_{r_0} := ((e/2)^\varepsilon, \beta')$; the assignment $(\alpha, \beta) \leftrightarrow (\alpha, \beta_{r_0})$ is a bijection between the support of $q_e^{B, (n)}$ and the support of $q_e^{B, (r_0)}$ (both consist of pairs where α has parts $< e/2$, β' has parts $< e/2$, and $|\alpha| + |\beta'| = r_0 - (e/2)\varepsilon$). Assemble:

$$\begin{aligned} q_e^{B, (n)} &= \sum_{(\alpha, \beta) \in \text{supp}(q_e^{B, (n)})} c_{\alpha, \beta}^{B, (n)} p_\alpha(x) p_\beta(y) && (\text{Thm 7}) \\ &= \sum_{(\alpha, \beta_{r_0}) \in \text{supp}(q_e^{B, (r_0)})} c_{\alpha, (e/2)^{2k'} \cup \beta_{r_0}}^{B, (n)} p_\alpha(x) p_{(e/2)^{2k'} \cup \beta_{r_0}}(y) && (\text{shape bijection}) \\ &= p_{(e/2)^{2k'}}(y) \sum_{(\alpha, \beta_{r_0}) \in \text{supp}(q_e^{B, (r_0)})} c_{\alpha, (e/2)^{2k'} \cup \beta_{r_0}}^{B, (n)} p_\alpha(x) p_{\beta_{r_0}}(y) && (\text{power-sum splitting}) \\ &= p_{(e/2)^{2k'}}(y) \sum_{(\alpha, \beta_{r_0}) \in \text{supp}(q_e^{B, (r_0)})} c_{\alpha, \beta_{r_0}}^{B, (r_0)} p_\alpha(x) p_{\beta_{r_0}}(y) && \text{by (12)} \\ &= p_{(e/2)^{2k'}}(y) \cdot q_e^{B, (r_0)} && (\text{Thm 7 at } r_0) \\ &= p_{(e/2, e/2)}(y)^{k'} \cdot q_e^{B, (r_0)}. && (p_{(e/2)^{2k'}} = p_{e/2}^{2k'} = p_{(e/2, e/2)}^{k'}) \end{aligned}$$

This proves the theorem. $\square$

6 Corollaries

Corollary 9 (n -independence of the type-B residue factor). $F_e^{B, (n)} := q_e^{B, (n)} / p_{(e/2, e/2)}(y)^{\lfloor n/e \rfloor}$ depends on n only through $r_0 = n \bmod e$; explicitly, $F_e^{B, (n)} = q_e^{B, (r_0)}$.

Proof. Immediate from Theorem 8. $\square$

Corollary 10 (Multiplicative recursion). For every even $e \geq 2$ and every $n \geq e$,

$$q_e^{B, (n)} = p_{(e/2, e/2)}(y) \cdot q_e^{B, (n-e)}.$$

Equivalently, $q_e^{B, (qe+r)} = p_{(e/2, e/2)}(y)^q q_e^{B, (r)}$ for $q \geq 0$ and $0 \leq r < e$.

Proof. Write $n = qe + r_0$ with $0 \leq r_0 < e$ and apply Theorem 8 to $q_e^{B,(n)}$ and $q_e^{B,(n-e)}$. $\square$

Corollary 11 (Fundamental domain, even e). *For every even $e \geq 2$, the sequence $(q_e^{B,(n)})_{n \geq 0}$ is the $p_{(e/2, e/2)}(y)$ -linear extension of its e “basic” values $q_e^{B,(0)}, q_e^{B,(1)}, \dots, q_e^{B,(e-1)}$.*

Proof. Direct from Corollary 10. $\square$

Corollary 12 (Type-B Ψ -companion identity). *For every even $e \geq 2$, $\Psi_{\emptyset, ((e/2)^2)}^B(\zeta_e) = 2e^2/(1 + \zeta_e^{e/2})^0 \cdot [\lim_{t \rightarrow \zeta_e} (1 - t^e)^2/(1 + t^{e/2})^2]$ evaluates cleanly to $2e^2$: the L’Hôpital limit in Lemma 3 at $M = 2$ gives $\lim_{m \rightarrow 1, 2} (1 - t^{em})/(1 + t^{e/2})^2 = 8 = 2^2 \cdot 2!$, and the “non-collapsed” numerator factors (indices $i \in [1, e]$ with $(e/2) \nmid i$) each fold via Lemma 2 into $(e/2)^2 = e^2/4$. Combining, $\Psi_{\emptyset, ((e/2)^2)}^B(\zeta_e) = 8 \cdot (e/2)^2 \cdot 1/1 = 2e^2$.*

Proof. Follows from the calculation in the proof of Proposition 6 at $(n, e, \alpha, \beta) = (e, e, \emptyset, ((e/2)^2))$ (so $k' = 1, r_0 = 0, \varepsilon = 0, M = 2, \alpha = \beta' = \emptyset$): equation (7) reads $\Psi_{\emptyset, ((e/2)^2)}^B(\zeta_e) = 2^2 \cdot 2! \cdot (e/2)^2 \cdot 1/1 = 8 \cdot e^2/4 = 2e^2$. $\square$

7 Verification

The identity $q_e^{B,(n)} = p_{(e/2, e/2)}(y)^{k'} q_e^{B,(r_0)}$ was verified with exact arithmetic in $\mathbb{Z}[\zeta_e]$ (no floating point) at the even- e test pairs of the discovery probe [3, 5]:

$$(n, e) \in \{(3, 2), (4, 2), (5, 2), (7, 2), (9, 2); (5, 4), (7, 4), (8, 4), (9, 4), (11, 4); (6, 6), (7, 6), (8, 6), (9, 6)\}.$$

Residuals are *exactly zero* on every test pair.

Sanity: three-route computation at $(n, e) = (8, 6)$, $(\alpha, \beta) = (\emptyset, (3, 3, 2))$. Here $k' = 1, r_0 = 2, \varepsilon = 0, M = 2, \beta' = (2), \alpha = \emptyset$.

Route 1: direct evaluation of $\Psi^B(\zeta_e)/z^B$. $\Psi_{\emptyset, (3, 3, 2)}^B(t) = \prod_{i=1}^8 (1 - t^{2i})/[(1 + t^3)^2(1 + t^2)]$. Set $u := t^3$; near $t = \zeta_e$ we have $u \rightarrow -1$ and $t^2 \rightarrow \zeta_3$. The numerator factors $1 - t^{2i}$ for $i = 3, 6$ contribute a double $(u + 1)^2$ zero which cancels $(1 + t^3)^2$. The residual computation gives $\Psi_{\emptyset, (3, 3, 2)}^B(\zeta_e) = 4(1 + u)^2 \cdot 2 \cdot [(1 - \zeta_3)(1 - \zeta_3^2)]^3/[(1 + u)^2(1 + \zeta_3)] = 216/(1 + \zeta_3) = 216e^{-i\pi/3} = 108 - 108i\sqrt{3}$; dividing by $z_{\emptyset, (3, 3, 2)}^B = 2^3 \cdot 3^2 \cdot 2! \cdot 2 = 288$ gives $(3/8) - (3/8)i\sqrt{3}$.

Route 2: closed-form (3). With $\alpha = \emptyset, \beta' = (2), h_1(2) = [1 \leq 2] + [1 + 3 \leq 2] = 1, h_2(2) = [2 \leq 2] + [2 + 3 \leq 2] = 1, \varepsilon = 0$: $c_{\emptyset, (3, 3, 2)}^{B, (8)} = 1 \cdot (1 - \zeta_e^2)(1 - \zeta_e^4)/[z_{\emptyset, (2)}^B \cdot (1 + \zeta_e^2)]$. Here $z_{\emptyset, (2)}^B = 2 \cdot 2 = 4$. And $(1 - \zeta_e^2)(1 - \zeta_e^4)/(1 + \zeta_e^2) = (1 - \zeta_e^2)^2(1 + \zeta_e^2)/(1 + \zeta_e^2) = (1 - \zeta_e^2)^2$ (using $\zeta_e^4 = \zeta_e^{-2}$ and $1 - z^2 = (1 - z)(1 + z)$ carefully: actually $(1 - \zeta_e^2)(1 - \zeta_e^4)$; with $\zeta_e^4 = \zeta_e^{-2}$ we have $(1 - \zeta_e^2)(1 - \zeta_e^{-2}) = 1 - \zeta_e^2 - \zeta_e^{-2} + 1 = 2 - (\zeta_e^2 + \zeta_e^{-2}) = 2 - 2\cos(2\pi/3) = 2 - 2(-1/2) = 3$; and $1 + \zeta_e^2 = 1 + \zeta_3 = e^{i\pi/3}$, so the ratio $3/e^{i\pi/3} = 3e^{-i\pi/3}$). Divide by $z^B = 4$: $c = (3/4)e^{-i\pi/3} = (3/8) - (3/8)i\sqrt{3}$.

Route 3: base coefficient $c_{\emptyset, (2)}^{B, (2)}$. $\Psi_{\emptyset, (2)}^B(t) = (1 - t^2)(1 - t^4)/(1 + t^2)$; at $t = \zeta_e$, this equals $(1 - \zeta_3)^2$ by the same simplification. Divide by $z_{\emptyset, (2)}^B = 4$: $c_{\emptyset, (2)}^{B, (2)} = (1 - \zeta_3)^2/4$. Compute: $|1 - \zeta_3|^2 = 3$ and $\arg(1 - \zeta_3) = -\pi/6$, so $(1 - \zeta_3)^2 = 3e^{-i\pi/3} = 3/2 - i3\sqrt{3}/2$; divide by 4: $(3/8) - (3/8)i\sqrt{3}$.

All three routes agree, confirming the coefficient identity (12) at literal $\mathbb{Z}[\zeta_e]$ equality.

Numerical sanity across ten support pairs (from cached probe [5]) confirms $c_{\alpha, \beta}^{B, (n)} = c_{\alpha, \beta_{r_0}}^{B, (r_0)}$ at test pairs $(n, e) \in \{(7, 4), (11, 4), (9, 6), (8, 6)\}$ with residuals bounded by 3×10^{-17} (floating-point noise).

8 Parity Dichotomy and the Verschiebung Projection

Combining Theorem 8 with the odd- e theorem [4] yields the following clean statement:

Corollary 13 (Type-B fundamental period, all e). *For every $e \geq 2$, $q_e^{B,(n)} = (p_e^B)^{\lfloor n/e \rfloor} \cdot q_e^{B,(n \bmod e)}$ where*

$$p_e^B = \begin{cases} p_{(e)}(x) & e \text{ odd}, \\ p_{(e/2,e/2)}(y) & e \text{ even}. \end{cases}$$

The mechanism is a two-alphabet “ $\mathbb{Z}/2$ -shadow” of Chevalley–Molien: the same tool (fake-degree = ratio of Weyl-numerator and Weyl-denominator, both a product of $(1 - t^\bullet)$ ’s) closes both cases, but the location of the collapse depends on parity. For odd e , the numerator has only $\lfloor n/e \rfloor$ zeros at $t = \zeta_e$; the denominator $\prod_a (1 - t^a)$ can match them (an e -cycle in α); and the β -side is inert since $\zeta_e^b \neq -1$ for any b . For even e , the numerator has $\lfloor 2n/e \rfloor$ zeros (twice as many, because $(e/2) \mid i$ is a weaker condition than $e \mid i$); the denominator $\prod_b (1 + t^b)$ can match them (an $(e/2)$ -part in β); and now the α -side is inert (see the calculation forcing $k_e(\alpha) = 0$ in Theorem 7).

Remark 14 (Fifth confirmation of the Chevalley–Molien-suffices pattern). The 2026-08-06 dream projected the Ayer–Kumari factorisation [6] of classical-group characters at odd powers of even roots of unity as the “right” modern tool for type-B self-similarity, with the reductive-group principal-torsion characters of Nishinaka–Prasad–Prasad [7] as a fallback. Neither is needed: the argument here is a two-page adaptation of the odd- e argument [4], using only Chevalley–Molien and *twin* classical cyclotomic identities (Lemmas 1 and 2). This is the *fifth* consecutive instance in the 2026-08 sprint (following: the support conjecture; type-A self-similarity; odd- e type-B self-similarity; the N-P-P read verdict) where a modern 2025 tool was projected as necessary and a classical calculation sufficed. Whether this pattern extends to the composite- d discussion—or breaks at a genuinely torsion-element or higher-cohomological phenomenon—is a question for the post-v1 arXiv methodological note.

Remark 15 (Two aesthetic points). (i) The prefactor collapse $2^M (e/2)^{2k'}/e^M = (2/e)^\varepsilon$ is where the “ $M = 2k' + \varepsilon$ ” bookkeeping earns its keep. The $(2/e)^{2k'}$ is exactly cancelled by the $2^{2k'}$ from the L’Hôpital limit and the $(e/2)^{2k'}$ from folding the non-collapsed numerator via Lemma 2; the residue $(2/e)^\varepsilon$ is exactly what appears in the base coefficient $c^{B,(r_0)}$ at parameter r_0 (with the same ε). The identity is k' -independent by construction.

(ii) The even- e “residue factor” $p_e^B = p_{(e/2,e/2)}(y) = p_{e/2}(y)^2$ is a single power sum on the negative alphabet, but appears *squared*. This is not an accident: the mass-bound analysis (9) forces *pairs* of $(e/2)$ -parts in β to appear together, because each $\zeta_e^{e/2} = -1$ contributes only *half* of the denominator zero (the negative side has zeros of order 1 per part, not 2; the numerator has zeros at every $(e/2) \mid i$, so 2 per period). The doubling is a $\mathbb{Z}/2$ -shadow, and $p_{(e/2,e/2)}$ is its geometric footprint.

References

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- [4] Clio, *Type-B self-similarity of the fake-degree Frobenius sum at a root of unity: The odd-e case*, `proofs/2026-08-08-type-b-self-similarity-odd-e.tex`.
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- [8] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Univ. Press, 1990.